REVATURE

Data Foundation - AI Engineer AWS -CTS -CAN

	MON	TUE	WED	THU	FRI	ENVIRONMENT
Week 1 **(Agile for Developers, Git Fundamentals, Python Fundamentals)**	Python-Fundamentals • Python-Orientation • Full Stack Overview • Interpreter vs Compiler • REPL Jupyter Agile for Developers • SDLC • Introduction to SDLC • Agile • Waterfall • Agile vs Waterfall • Story pointing • Scrum Ceremonies Git Fundamentals • Git Fundamentals • Initializing A Repository • Pushing To A	Project 0 Python - Fundamentals Python-Basics • Python Syntax • Comments • Variables and Datatypes • Operators • User Input and Output • What is Python Why Python Professionalism at Rev	Project 0 Python-DataTypes • Strings • Casting • Boolean • Lists • Tuples • Range • Sets • Binary Type • Nontype • Dictionaries • Numbers • Datetime • Namespaces	Project 0 Python-FlowControls-Stmts • If-Else • While • For Python-Functions & Arrays • Function • Lambda • Arrays	Project 0 Python-Classes & Inheritance • OOP Concepts • Inheritance • Iterators • Scope • Classes and Objects Feedback Form	

	MON	TUE	WED	THU	FRI	ENVIRONMENT
	Remote Repository • Git Commit, Branch, Merge, Push, Pull • Source Control Management(git,vcs,cvcs,dvcs) Git Exercise					
Week 2 **(Python Advanced)**	Project 0 Python-Modules • Math • Logging • JSON • Regex • pip and install pip • pylint • Module	Project 0 Python-Exception Handling • Error • Exception Handling • try-except Python- File Handling • File Handling • Read Files • Write Create Files • Delete Files	Project 0 Testing Overview • Unit Testing • Pytest Overview of Data Visualization • Matplotlib • Pandas • Numpy Interview Mindset - 1st Part	Project 0 • Counters • Named Tuples • Ordered Dicts • Collections Module	Project 0 Python Review Feedback Form	
Week 3 **SQL**	Project 0 MySQL SQL-Introduction • Introduction To RDBMS	Project 0 Sub-Languages • DDL • DML • DQL	Project 0 Advanced-Schema • Primary Key • Composite Key • Foreign Key	Project 0 • Clauses • Scalar Functions • Aggregate Functions	Project 0 Advanced - SQL • Trigger • Views • What Is A Stored	

	MON	TUE	WED	THU	FRI	ENVIRONMENT
	• What Is SQL • Schema • Table Structure • SQL Data Types • Data Modeling And ERD • Defining Schema	• DCL • TCL • Overview Of Sublanguages Getting Ready for Your Interview -2nd Part	• Referential Integrity • Unique Key • Secondary Alternate Key • Normalization • Multiplicity • Consistency • CREATE DROP TRUNCATE • Constraints • Auto Incrementing • CHECK • DEFAULT • CASCADE • INSERT • UPDATE • DELETE	Sub-Queries & Joins • What Is A Subquery • What Is A Join • Aliases • Cross Join • Equi And Theta Joins • Inner Join • Left And Right Joins • Outer Join • Transaction Commit Rollback Isolation Levels • What Is A Transaction • ACID Properties • Transaction Properties • CRUD Operations • Sequence	Procedure • What Is A User Defined Function • Indexes Review Feedback Form	
Week 4 **Data Fundamentals, Cloud Concepts, Data Warehousing, ELT/ETL Concepts**	Project 0 • Cloud Computing Model Types • Cloud Computing Service Types	Project 0 • Big Data Fundamentals • Components of Big Data	Project 0 • Data Warehousing Introduction • Data Store Vendors	Project 0 • Modeling • Conceptual/Logical/Physical	Project 0 • ETL overview • Bigquery Connections	

	MON	TUE	WED	THU	FRI	ENVIRONMENT
	• Cloud Computing Definition • IaaS, PaaS, SaaS • Challenges • Pricing • AWS/Azure/GCP (Data Services)	• Architecture • Benefits • Challenges • Data Lifecycle Stages • Types of Data: Structured, Semi-structured, Unstructured • File Types	• OLTP versus OLAP • Data Warehouse Architecture • Data Warehouse versus Datalake • Operational Data Store • Data Mart • Data Cleansing • Bigquery • Creating Datasets • Create and Use Tables • Bigquery Queries • Advanced Queries • Denormalization	• Dimension Modeling • Star Schema • Snowflake Schema • Slowly Changing Dimensions • SCD Types	• Building ETL/ELT Processes Review Feedback Form	

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Week 5 **Gen AI Introduction**	Project 0 AI Orientation • ML and DL Introduction • GenAI Overview • AI Introduction Prompt Engineering • Zero-Shot Prompting • Few-Shot Prompting • Constraints • Fine-Tuning And Conditioning • Interation And Dialog State • Instructions And Guidelines • Hallucinations • Responsible Usage • Security • Prompt Engineering Introduction	Project 0 LLM • LLMs (GPT, BERT, Claude, Llama, Copilot, Codeium) • Use Cases For LLM • LLM Best Practices • LLM Security Considerations	Project 0 AI tooling • Gen AI for developers • Introduction to AI pair programming • Overview of Copilot • Integration with IDE • Using Gen AI for code generation • Using Gen AI for documentation	Project 0 • Code optimization • Code analysis • Responsible use • Security risks • AI tooling review	Feedback Form	
Week 6 **AI/ML fundamentals**	• Intro to Machine Learning - algorithms and use cases	Unsupervised learning - clustering • Distance between	Project 1 Intro to Deep Learning	Project 1 Tensorflow Introduction	Project 1 Review Topics	

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	Supervised/unsupervised learning • Applications of regression • Types of regression, fitting the line • Applications of classification	clusters • k-means algorithm	• Neural networks - types • CNN & Image classification • Artificial neural network, layers • Natural Language Processing - techniques, components, applications	History of Tensorflow • Implementing basic graphs • Reducing tensors of arbitrary shape • Graphs, sessions, TensorBoard	• Activation functions • Perceptrons - illustration, training, parameters Tensorflow basics • Graph visualization • Constants, placeholders, variables • Creating a model Weekly Survey Feedback	
Week 7 **Tensorflow**	Project 1 • Limitations of single perceptrons • Multi-layer perceptrons • Backpropagation algorithm • Understand backprop - neural network	Project 1 • MLP digit classifier using tensorflow • TensorBoard • Collaborative filtering with RBM • Autoencoders	Project 1 TFlearn API • Composing models in TFlearn • Sequential composition	Project 1 • Functional composition • Predefined neural network layers • Batch normalization • Saving and loading a model with TFlearn	Project 1 • Customizing the traning process • TensorBoard with TFlearn Weekly Survey Feedback	
Week 8 - Vector Databases **(Feature Engineering, VectorDB)**	Project 1 Feature Engineering • Encoding categorical data • Handling missing	Project 1 • Model Evaluation & Metrics • Accuracy, Precision, Recall, F1-score	VectorDB VectorDB-Orientation • VectorDB Introduction	Project Review VectorDB-Database-Management • Index Management	VectorDB-Semantic-Search • Semantic Search • Data Preprocessing	

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	values, outliers • Feature selection techniques	• Confusion matrix interpretation • Avoiding overfitting (regularization, cross-validation)	• Vector Overview • Vector Database Use Cases • Storing and Retrieving Data in SQLite • Vector Similarity Search VectorDB-Orientation • Chroma DB Setup • Local Vector DB with Chroma DB • Pinecone Overview • CRUD with Vector Data • Pinecone Setup	• Collection • Partitioning Vectors • Upsert • Vector Partitioning Using Metadata • Distance Metrics	• Preparing for Upsert • Vector Query with Semantic Search • Named Entity Recognition (NER) • Vector Embedding Models • Data Preparation • NER Helper Function • Vector Embedding in Batches • Metadata Processing VectorDB Review Weekly_Survey Feedback	
Week 9 - LangChain (LangChain)	LangChain LangChain-Orientation • Introduction to Retrieval-Augmented Generation (RAG) • Introduction to Models • Using LLMs with LangChain examples	LangChain-Connections • Introduction to Data Connections • Document Loaders • Document Transformers • Text Embedding • Vector Store	Project 2 LangChain-Chains • SequentialChain • LLMRouterChain • TransformChain • MathChain • Structured output design & audit-trail	Project 2 LangChain-Memory • Introduction to Memory • ChatMessageHistory • ConversationBufferMemory	Project 2 LangChain Integrations • StreamLit • Adding to LangChain OpenSource Project LangChain Review Weekly_Survey Feedback	

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	• Bedrock Fundamentals • Overview of Amazon Bedrock architecture • API usage via AWS SDK • Connect Bedrock LLMs to LangChain LangChain-Prompts • Prompt Templates • Parsing Output • Serialization of Prompts	• Retrievers • Multi-query Retrieval LangChain-Chains • LangChain Chains • LLMChain • SimpleSequentialChain	logging for reliability	• ConversationBufferWindowMemory • ConversationSummaryMemory		
Week 10 **LangGraph**	Project 2 • What is "Agentic AI"? Difference between LLMs, chains, and agents • Core components: autonomy, planning, memory, and tool-use • Why traditional chains fall short for complex reasoning • The evolution from LangChain to	Project 2 LangGraph Overview • LangGraph Fundamentals • LangGraph architecture: graphs, nodes, edges, states, and reducers • Creating multi-step workflows with branching logic • Managing memory and async tool	Project 2 • How multiple AI agents coordinate (planner–executor–critic pattern) • Communication protocols between agents • Long-term memory and shared state stores • Tools integration (APIs, databases, vector stores)	Project 2 Introduction to crewAI • Open-source framework for collaborative agent systems • multi-role, multi-agent orchestration • Integration with LangChain and LangGraph • Agent identity & session security	Project 2 LangGraph Review Weekly_Survey. Feedback	

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	LangGraph • Human-in-the-loop design	execution • Comparison to LangChain Agents	• Agent decomposition rationale • Choosing a multi-agent architecture pattern • Pipeline/sequential pattern • Supervisor/router pattern (hub-and-spoke) • Hierarchical pattern • Network/peer-to-peer pattern • Parallel fan-out/fan-in pattern			
Week 11 **(AWS Sagemaker, MLOps)**	Project 2 Vertex AI Sagemaker Orientation • Introduction to AWS SageMaker and managed ML services • SageMaker Studio setup and IAM role permissions • Project structure and dataset preparation for	Project 2 Model Training & Deployment • Preparing TensorFlow models for deployment (SavedModel, .tar.gz) • Using SageMaker Estimators and the Python SDK for training jobs • Creating real-time	Project 2 Model registry & monitoring • Tracking experiments with SageMaker Experiments • Managing model versions with the Model Registry • Endpoint deployment and	Project 2 SageMaker Pipelines & CI/CD Automation • Introduction to SageMaker Pipelines for MLOps automation • Creating pipeline steps: Data Prep/ Train/Evaluate/Registe r/ Deploy • Integrating Model Registry with	Project 2 MLOps Orientation Weekly_Survey. Feedback	

	MON	TUE	WED	THU	FRI	ENVIRONMENT
	ML • Overview of the SageMaker lifecycle (data /train /deploy /monitor)	and serverless inference endpoints • Testing predictions and optimizing performance	rollback strategies • Proactive monitoring pattern	deployment workflows		

PROJECT	TECHNOLOGIES
Project 1	Python, Tensorflow
Project 2	Python, Git, Agile, LLMs, LangChain, RAG, VectorDB, LangGraph
Project 0	

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